



"Anything done manually twice gets automated"

Nikita Petrich

FREELANCE SENIOR FULL-STACK & AI ENGINEER

Phone: +49 15679088678 **Mail:** n.petrich@sequenz.io **Web:** sequenz.io

Location: Remote (preferred) · Munich hybrid: 1–2 days/week · Long-distance travel: 1–2 days/month (if needed)

Onboarding: on-site at the start (if needed)

Availability: immediately available · full-time (full capacity) **Rate:** on request

Qualification: IHK-certified Application Developer (Fachinformatiker, Cologne)

Experience: 7+ years

FOCUS AREAS

LLM integration RAG GDPR-compliant AI architecture AI-powered automation
 Agentic coding / AI engineering TypeScript Python Backend (NestJS · Node.js · FastAPI)
 Frontend (Next.js · React · Angular) PostgreSQL Clean Architecture Microservices CI/CD
 Docker

LANGUAGES

German · Native English · B2 · US/EU remote-team experience

WAY OF WORKING

results-oriented self-driven & reliable autonomous, asynchronous way of working
 clear communication (EN/DE)

PROFILES

[Website](#) [LinkedIn](#) [GitHub](#) [freelancemap](#) [Malt](#) [Certificates](#)

① I'm a freelance Senior Full-Stack & AI Engineer focused on AI: **LLM integration, RAG and AI-driven automation** – from architecture all the way to stable production.

My guiding principle: **anything done manually twice gets automated**. That applies to my clients' processes as much as to my own work. In logistics, a route management system I built saves **up to 1,000 hours a year** through automated document workflows; in retail, **up to 40 hours of admin work per month**; at a notary's office, client phone enquiries dropped by **up to 70 %**.

I apply the same logic to development itself. I work consistently with **agentic coding and AI engineering workflows** (Claude Code, Cursor, Code Rabbit, MCP) – spec-driven, with AI-assisted test generation and agentic review processes. This isn't a gimmick, it's the reason for unusual speed at consistent quality: a complete cross-platform app from first commit to handover **in roughly two months**, as the sole developer.

How I work: **independently, remote and asynchronously**. I don't need daily direction, just a clear goal – I'll figure out the path there myself and speak up before anything slips. I document decisions, ship tested and review-ready code, and flag early when a requirement turns out to be technically more expensive than expected. I communicate in German and English.

Technically: **TypeScript and Python**, backends with NestJS, Node.js and FastAPI, frontends with Next.js, React and Angular, plus PostgreSQL, Docker, CI/CD and cloud on Azure or GCP. Clean Architecture and microservices are standard; **GDPR compliance and Secure by Design** are built in from day one.

I've done this inside a **20-person engineering team** – most recently on an AI-powered LegalTech platform with **more than 3,000 customers** – and as a sole developer with full product ownership. Over **seven years** across LegalTech, HealthTech, e-commerce, EdTech and logistics.

Does this fit what you have in mind? Book a [free intro call](#)

Skills & Expertise

Complete overview of my technical skills

01 AI, LLM & AI ENGINEERING

LLM Integration RAG (Retrieval Augmented Generation)
 Prompt Engineering Embeddings Vector Databases pgvector
 Semantic Search Document Analysis OCR LangChain
 OpenAI API Azure OpenAI Anthropic Claude API Hugging Face
 Ollama AI Agents Agentic Software Engineering
 AI-Powered Automation AI Engineering Artificial Intelligence
 GDPR-Compliant AI Architecture

03 PROGRAMMING LANGUAGES

TypeScript JavaScript Python C# SQL Bash HTML
 CSS SCSS XML UML

05 FRONTEND

React Next.js Angular React Native Ionic Capacitor
 Cross-Platform Development iOS Android Redux NgRx
 RxJS TanStack Query React Hook Form React Router Axios
 Zod Tailwind CSS shadcn/ui Material UI (MUI) Fluent UI
 Single Page Application (SPA) Progressive Web App (PWA)
 Server-Side Rendering (SSR) Internationalization (i18n)
 Frontend Development Web Development

07 DATABASES & DATA

PostgreSQL MySQL MariaDB Microsoft SQL Server (MSSQL)
 MongoDB Mongoose NoSQL SQLite Redis Cloud Firestore
 Prisma MikroORM TypeORM Drizzle ORM
 Entity Framework Core Data Modeling Data Architecture DTOs
 Data Synchronization ETL CSV Processing Caching
 Idempotency

09 AUTH & SECURITY

JWT OAuth 2.0 Single Sign-On (SSO) better-auth
 Firebase Authentication (Auth) Firebase Security Rules
 Authentication Role-Based Access Control (RBAC)
 Row Level Security SSL/TLS End-to-End Encryption
 O365 Identity Microsoft Entra ID (formerly Azure Active Directory)
 Secure by Design IT Security

02 AGENTIC CODING

Claude Code Cursor AI GitHub Copilot Code Rabbit ChatGPT
 Lovable v0 Claude Code Hooks Slash Commands Agent Skills
 Subagents Multi-Agent Workflows Model Context Protocol (MCP)
 Agentic Coding Context Engineering Spec-Driven Development
 Agentic Pull-Request Workflows AI-Powered Test Generation

04 BACKEND & FRAMEWORKS

Node.js NestJS Express.js Fastify FastAPI Flask
 ASP.NET Core .NET Core GraphQL REST API OpenAPI
 Swagger WebSockets Webhooks Message Queues BullMQ
 RabbitMQ Backend Development
 Object-Oriented Programming (OOP)

06 DESIGN & UX

Figma Wireframing Prototyping Design System Design Tokens
 Component Libraries Storybook UI/UX Design UI Implementation
 Responsive Design Mobile First Accessibility (WCAG) Usability
 User Flows Onboarding Flows Information Architecture
 UX Writing Microcopy Design Reviews

08 DEVOPS, CLOUD & INFRASTRUCTURE

Docker Docker Compose CI/CD Continuous Integration
 GitHub Actions GitLab CI Azure Pipelines Jenkins Nginx
 Sentry Monitoring Observability Linux Ubuntu
 Self-Hosting Microsoft Azure Azure Functions Azure App Service
 Azure Blob Storage Azure SQL Google Cloud Platform (GCP)
 Amazon Web Services (AWS) Hetzner IONOS Digital Ocean
 Vercel Supabase Firebase Cloud Functions Cloud Storage
 Firebase Cloud Messaging (FCM) Firebase Hosting DevOps
 DevSecOps

Skills & Expertise

Continued — Methodology, Compliance, Tooling & Domains

10 ARCHITECTURE & PRINCIPLES

Software Architecture Clean Architecture Clean Code SOLID
Design Patterns Domain-Driven Design Separation of Concerns
Dependency Injection Microservices Event-Driven Architecture
Monorepo Nx Turborepo Scalable Architecture
System Integration Platform Migration Middleware API-First
Resilience Patterns Type Safety MVP Development

12 BUSINESS & STRATEGY

IT Strategy Digital Strategy Digital Transformation
Process Digitization Product Ownership Entrepreneurship

14 TESTING & QA

Jest Vitest Cypress Playwright React Testing Library
Unit Testing Integration Testing End-to-End Testing Postman
API Testing Test Automation

16 DOMAINS & INDUSTRIES

LegalTech GovTech Law Firm Software Notary Services
Client Portal Document Management HealthTech
Medical Technology Maintenance Management ERP
Inventory Management Warehouse Management Invoicing
Order Management CRM Accounting Master Data Management
Multi-Tenancy Stripe SEPA Payment Processing Logistics
Route Planning PDF Generation DOCX Generation
Digital Archiving EdTech E-Learning
Learning Management System (LMS) IoT Device Pairing
Consumer Electronics Social Media SaaS Platform Development

11 METHODOLOGY & COLLABORATION

Agile Methods Scrum Kanban Scrumban Code Reviews
Test-Driven Development (TDD) Software Development
Project Management IT Consulting Requirements Engineering
Remote Work Public Speaking

13 DATA PRIVACY & COMPLIANCE

GDPR Privacy by Design Data Minimization
Data Processing Agreement (DPA) EU AI Act AI Governance GoBD
Audit-Proof Record-Keeping EU Hosting Digital Sovereignty
German Accessibility Act (BFSG) European Accessibility Act (EAA)

15 TOOLING

Git GitHub GitLab Husky Visual Studio Code Visual Studio
Jira Linear Azure DevOps Bluedot HubSpot Cal.com
Slack Google Workspace (Meet, APIs) Amplitude
Product Analytics

17 ROLES & PROFILE

Senior Full-Stack Engineer AI Engineer Software Engineer
Freelance Developer Interim CTO Technical Consultant
Certified Application Developer (Fachinformatiker) German (Native)
English B2 Munich DACH

Project Overview

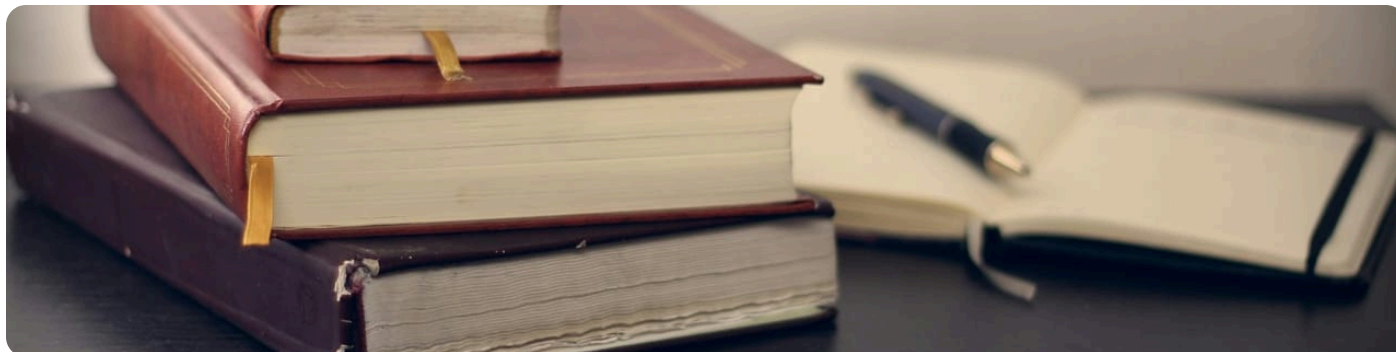
All projects at a glance — with direct links to each detail page

01	BESCHEIDKLAR · Own Product Founder & CTO	LegalTech / GovTech	04/2025 – present ~15 months
02	Manifest OS · Manifest Law Full-Stack (AI features)	LegalTech / Immigration	12/2025 – 04/2026 ~5 months
03	RateUp · Krutzfeld Tech GmbH Sole Developer · Full-Stack	Social Media / Mobile App	11/2025 – 12/2025 ~2 months
04	AITOI · AITOI Frontend Engineer	IoT / Consumer Electronics	08/2025 – 09/2025 ~2 months
05	DiNo · LeXtorByte UG Frontend	LegalTech / Notary	01/2025 – 05/2025 ~5 months
06	AccountingOS · HD Autoservice Sole Developer	Retail & Automotive (Inventory Mgmt.)	01/2024 – 12/2024 ~12 months
07	Ladetrans · CDH Stephanus Sole Developer	Logistics & Transport	02/2021 – 05/2023 ~2 years 4 months
08	XU Navigator · XU Group Full-Stack	EdTech / E-Learning	05/2020 – 02/2023 ~2 years 10 months
09	Medical Device Management System · Hospital Group (anonymized) Full-Stack	HealthTech / Medical Devices	01/2019 – 05/2020 ~1 year 5 months

LEGALTECH / GOVTECH

01

04/2025 – PRESENT · 15 MONTHS

BESCHEIDKLAR — AI-Powered LegalTech SaaS Platform**Role:** Founder & CTO **Client:** Own Product**Team:** Solo (sole developer) · own product (side project)**Location:** Lower Saxony, Germany · Remote**Project Language:** German**Website:** bescheidklar.de**Methodology:** Self-directed · direct product ownership**RESPONSIBILITIES**

- Designed and built a two-sided LegalTech SaaS platform with a modular, service-oriented architecture — combining AI-powered document analysis (**Azure OpenAI**) with a regional licensing model for law firms.
- Designed and implemented a multi-stage **AI pre-screening pipeline** for official notices and terminations (in progress): extraction → deadline check → plausibility and success assessment, with structured case preparation.
- Automated, GDPR-compliant handoff of qualified leads (incl. uploaded/photographed documents, calculations, and AI results) directly into the system of matching specialists (lawyers, law clinics) — including contact details, producing an immediately actionable case lead.
- Built an AI-powered advice section with **RAG**-based content generation — articles and recommended actions for affected individuals (e.g. after termination or a rejection notice).
- Covered multiple legal domains (legal, social, and labor law) via an extensible, multilingual, low-barrier web interface.
- Full ownership of architecture, tech stack, EU-infrastructure hosting, and a secure-by-design compliance concept (**EU AI Act**, **GDPR**, **German Legal Services Act**).
- Built type-safe end-to-end data flows (**TypeScript**, **DTOs**, **Zod**) with automated **CI/CD** delivery (**GitHub Actions**, **Docker**), monitoring/error tracking (**Sentry**), and test coverage (**Vitest**, **Playwright**).

RESULTS

- MVP delivered single-handedly — architecture, platform and operation on EU infrastructure from one source.
- Compliance built into the product design rather than retrofitted: the EU AI Act, the GDPR and the limits of the German Legal Services Act were part of the plan from the start.
- Addressable market: **167,547** admitted lawyers in Germany (German Federal Bar, 1 Jan 2026); scalable model with minimal fixed costs.
- Target: up to **10,000** automated pre-screenings per month via AI-powered initial assessment.
- Target: up to **80 %** less manual case intake through AI intake and automated lawyer routing.
- Target: predictable inflow of pre-qualified cases for partner law firms instead of costly manual initial assessment.

TECHNOLOGIES

TypeScript Python Next.js React FastAPI LLM Integration OpenAI API Azure OpenAI RAG (Retrieval Augmented Generation)
 Prompt Engineering Embeddings pgvector OCR Document Analysis LangChain PostgreSQL Redis Tailwind CSS shadcn/ui
 Stripe JWT OAuth 2.0 Docker GitHub Actions CI/CD Hetzner Linux Ubuntu Slack Husky Microservices
 Clean Architecture SaaS EU AI Act GDPR Privacy by Design Product Ownership

LEGALTECH / IMMIGRATION

02

12/2025 – 04/2026 · 5 MONTHS

Manifest OS — AI-Powered Immigration Platform

Role: Full-Stack Engineer (AI Feature Integration)**Client:** [Manifest Law](https://manifestlaw.com)[↗ Recommendation](#)**Team:** ~20 engineers across 4 teams (platform, AI, full-stack) · cross-functional collaboration with product, design, and tech leads**Location:** New York, USA · Remote**Project Language:** English**Website:** manifestlaw.com**Methodology:** Scrum · autonomous way of working

RESPONSIBILITIES

- Independently implemented production AI features: event-driven enrichment of attorney intake-call leads — meeting recordings are transcribed via **Bluedot** and processed resiliently (retry, idempotency) through a webhook, a **BullMQ** queue, and the **OpenAI** integration; structured results flow back automatically into the core systems.
- AI-powered evidence processing: uploaded case documents (e.g. ID scans, birth certificates, proofs) are automatically analyzed, classified, and consistently named.
- Contributed to consolidating four separate portals (B2B, B2C, Lawyer, Ops) and one API into a unified “B2X” portal — in a team of four engineers, owning the full-stack migration of the B2B and Ops areas, including merging the data model and UI.
- Introduced app-wide event tracking (**Amplitude**) to analyze user behavior — identifying weak points and friction as a data foundation for product and data analytics.
- Built resilient, event-driven backend services (**NestJS**, **BullMQ**) with retry, idempotency, and error-handling strategies for reliable background processing.
- Delivered within a monorepo with clearly scoped feature modules to enable a fast future split into microservices, with type-safe interfaces (**TypeScript**, **DTOs**) and automated tests (**Jest**, **Playwright**) in the CI/CD pipeline.
- Worked closely in code reviews and used AI-powered tools (**Claude Code**, **Cursor AI**, **Code Rabbit**) to safeguard quality and development speed.

RESULTS

- Part of a roughly 20-person engineering team on an established platform (company valuation in the range of ~\$60M USD).
- AI features running in production: intake-call leads and case documents are enriched, classified and named without manual rework.
- Application-wide event tracking established as a shared data foundation for product and data analytics.
- Refactoring initiatives in the migrated B2B and Ops areas: existing structures simplified and geared towards better maintainability.
- Platform with **over 3,000 customers**, **150+** corporate programs and more than 100 lawyers (team result).
- Up to **15 %** higher approval rates versus the USCIS average (team result).

TECHNOLOGIES

TypeScript Python Next.js React NestJS Node.js Fastify Express.js REST API Webhooks DTOs BullMQ RabbitMQ
 LLM Integration OpenAI API Anthropic Claude API Model Context Protocol (MCP) Bluedot HubSpot Cal.com Amplitude PostgreSQL
 Redis MikroORM TanStack Query better-auth JWT Docker Linux Ubuntu Google Cloud Platform (GCP) GitHub Actions CI/CD
 Claude Code Cursor AI Code Rabbit Linear Figma Slack Playwright Husky Microservices Event-Driven Architecture
 Idempotency Clean Architecture

SOCIAL MEDIA / MOBILE APP

03

11/2025 – 12/2025 · 2 MONTHS

RateUp — Rating App for Web, iOS and Android

Role: Sole Developer · Full-Stack **Client:** [Krutzfeld Tech GmbH](#)**Team:** Solo (sole developer) · client project**Location:** Zurich, Switzerland · Remote**Project Language:** German**Website:** krutzfeld.tech**Methodology:** Self-directed

RESPONSIBILITIES

- Designed and built a social-media rating app (feed, ratings, profiles) from a single **Ionic/Angular** codebase — as an installable **PWA** on the web and as native builds for **iOS** and **Android (Capacitor)**.
- Registration, login, and onboarding incl. profile creation via **Firebase Authentication**, secured with **Firebase Security Rules**.
- Built the social feed with filters, user posts, and media upload (**Cloud Firestore**, **Cloud Storage**).
- Server-side logic in **Cloud Functions**, with delivery via **Firebase Hosting** and a **CI/CD** pipeline (**GitHub Actions**) for all three platforms.
- Built a maintainable, handover-ready codebase: clearly layered structure, reusable components, type-safe models, unit tests, and consistent clean-code style (**SOLID**).

RESULTS

- Delivered a finished product to the client — running as a PWA on the web and as a native app on iOS and Android from a single shared codebase.
- Fully built on **Firebase**: authentication, data storage, media storage, server logic, and delivery.
- Handover-ready codebase: consistent structure, reusable components, unit tests, and full type safety — the client team can continue without an onboarding hurdle.
- From first line of code to handover **in around two months**.

TECHNOLOGIES

TypeScript Angular Ionic Capacitor RxJS Cross-Platform Development iOS Android Progressive Web App (PWA)

Responsive Design UI Implementation User Flows Firebase Firebase Authentication (Auth) Firebase Security Rules Cloud Firestore

Cloud Storage Cloud Functions Firebase Cloud Messaging (FCM) Firebase Hosting Authentication Onboarding Flows Social Media

Type Safety Unit Testing Clean Code SOLID Separation of Concerns Clean Architecture Git GitHub GitHub Actions CI/CD

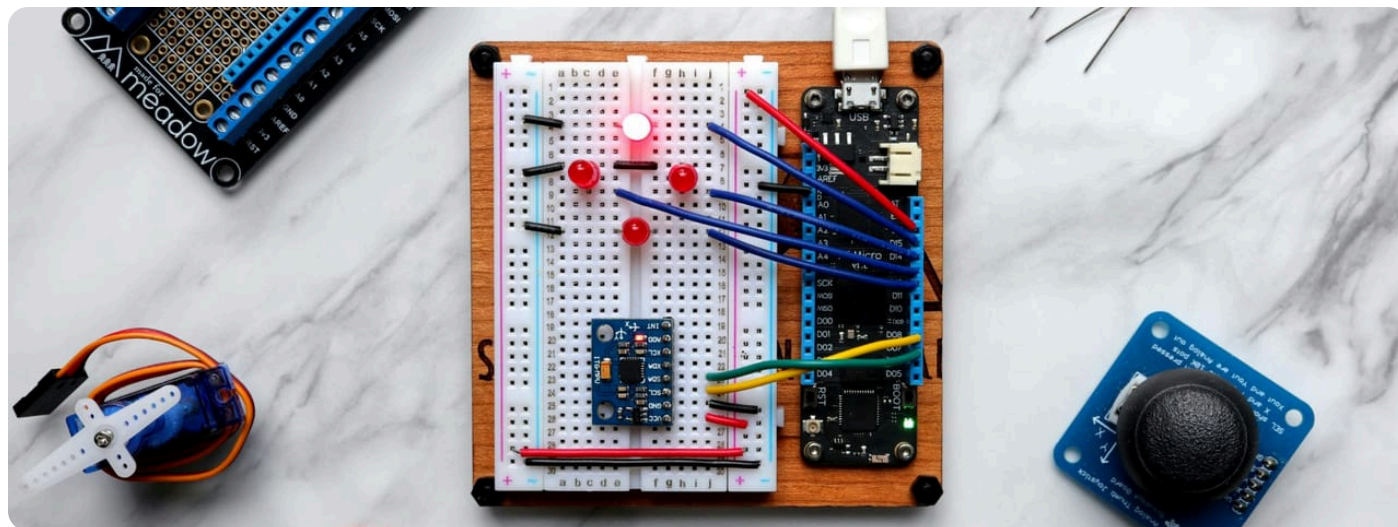
Linux Trello

IOT / CONSUMER ELECTRONICS

04

08/2025 – 09/2025 · 2 MONTHS

AITOI — Interactive IoT Toy (MVP)

Role: Frontend Engineer**Client:** AITOI[➤ Recommendation](#)**Team:** Small cross-disciplinary team incl. hardware development**Location:** Frankfurt, Germany · Remote**Project Language:** German & English**Visibility:** internal system (code not public)**Methodology:** Self-directed · English-language coordination with hardware team

RESPONSIBILITIES

- Designed and built the IoT Wi-Fi onboarding for a connected AI toy via QR-code scan — incl. a guided setup flow, device pairing, and robust error handling.
- Implemented first-time device setup as well as later reconfiguration of settings via a platform-independent **Progressive Web App (PWA)**.
- PWA dashboard for parents: follow, filter and review their children's play sessions.
- Integrated **Supabase** for authentication, data storage, and real-time sync between device and app (incl. **row level security**).
- Built a performant, platform-independent PWA architecture (**Next.js**) with a responsive UI.
- Defined and implemented type-safe data models and API interfaces (**TypeScript**, **DTOs**, **Zod**).
- Coordinated closely, in English, with the hardware team to define device-app communication and the setup protocols.
- Advised and guided the client from initial idea to an actionable **MVP** concept.

RESULTS

- Fully functional **MVP** frontend in **~2–3 weeks of actual working time** — taken independently from concept through client coordination to the demo.
- Setup app and parent dashboard as one platform-independent PWA with real-time sync between the IoT device and the app.
- Scalable, documented architecture as a solid technical foundation for the product launch.
- Trusted, cross-functional collaboration with the hardware team throughout the project — despite the tight timeline, no friction at the device-app interface.

TECHNOLOGIES

TypeScript Next.js React Progressive Web App (PWA) Responsive Design Design System React Hook Form DTOs Device Pairing
 Supabase Supabase Realtime Row Level Security PostgreSQL Tailwind CSS shadcn/ui JWT Authentication Docker Linux
 Ubuntu CI/CD GitHub Actions Vitest Cursor AI Model Context Protocol (MCP) MVP Development Clean Architecture

LEGALTECH / NOTARY

05

01/2025 – 05/2025 · 5 MONTHS

DiNo — Digital Notary**Role:** Frontend Engineer**Client:** [LeXtorByte UG](#)[↗ Recommendation](#)**Team:** 2–4 people**Location:** Oranienburg, Germany · Remote**Project Language:** German**Website:** digitales-notariat.de**Methodology:** Scrumban**RESPONSIBILITIES**

- Implemented frontend features to digitize notary processes with a focus on clear, traceable user flows — incl. client portal and notary portal.
- Built digital case management: structured data capture, digital case files, and seamless internal filing to speed up case processing.
- Contributed to extending a client portal for status tracking, secure document delivery, and privacy-compliant communication.
- Built reusable components and unified form validation (**React Hook Form**) to speed up feature development across the team.
- Initiative within the codebase: refactored existing code and fixed long-standing bugs — aimed at solutions that hold up over time.
- Coordinated closely with design and backend in a **Scrumban** rhythm to quickly translate new requirements into usable interfaces.

RESULTS

- Per the provider, up to **70 %** fewer client phone inquiries.
- Per the provider, invoices are paid up to **30 days** sooner.
- Noticeably reduced administrative overhead and greater transparency through consistently digital, traceable processes.
- Consistent, modern UI across the client and notary portals, strengthening both client trust and notary-office efficiency.

TECHNOLOGIES

TypeScript React Python Flask REST API OpenAPI DTOs MariaDB MySQL Redux React Router React Hook Form
 Material UI (MUI) Single Page Application (SPA) User Flows Client Portal Digital Archiving JWT OAuth 2.0
 Role-Based Access Control (RBAC) GDPR Docker Nginx Linux Ubuntu GitLab Slack Clean Architecture

RETAIL & AUTOMOTIVE (INVENTORY MGMT.)

06

01/2024 – 12/2024 · 12 MONTHS · ONGOING MAINTENANCE (SIDE PROJECT)

AccountingOS — GoBD-Compliant Accounting & Inventory System

Role: Sole Developer · Full-Stack & AI**Client:** HD Autoservice[➤ Recommendation](#)**Team:** Solo (sole developer) · client project**Location:** Salzgitter, Germany · Remote**Project Language:** German**Visibility:** internal system (code not public)**Methodology:** Self-directed

RESPONSIBILITIES

- Designed and built a feature-rich, **GoBD**-compliant accounting and inventory management system on top of a cleanly layered architecture (**Clean Architecture**).
- Designed and implemented a resilient data-sync service (“pipe service”) that processes supplier catalogs (auto parts) with up to **20M** CSV rows over a TLS-encrypted connection — with queuing, caching, retry logic, validation, deduplication, and data cleansing.
- Built the accounting module: invoices, credit notes, internal orders, PDF export, customer management, **inventory management**, order import with automatic order conversion, and **GoBD** record-locking.
- Coupled **inventory management** with invoice management: automatic stock deduction on invoice line items and notification when falling below minimum stock.
- Recursive resolution of foreign-key dependencies in input dialogs for consistent master-data maintenance.
- Customer and **master-data management** with role-based access control (admin, staff, tax advisor) and authentication via **better-auth**.
- Advised the client on technical approaches, architecture, and process optimization.
- Built type-safe **GraphQL** and REST interfaces (**NestJS**) with validation, error handling, and consistent data modeling (**MikroORM**, **DTOs**).
- Ongoing support (side project): maintenance, security updates, and further development.

RESULTS

- According to the client, saves up to **40 hours** of admin work per month (up to **480 hours/year**) — about **€ 19,200** a year at an assumed labour cost of € 40 per hour.
- Cuts time spent on routine processes like invoicing and **inventory management** by up to **50 %**.
- Consistently digital, **GoBD**-compliant workflows with direct coupling of inventory and **accounting**.

TECHNOLOGIES

TypeScript Node.js NestJS Next.js React Fastify REST API GraphQL DTOs Type Safety Data Synchronization
 CSV Processing Message Queues Caching MariaDB PostgreSQL Redis MikroORM TanStack Query Tailwind CSS
 Inventory Management Warehouse Management Accounting ERP GoBD better-auth JWT Role-Based Access Control (RBAC)
 Docker Self-Hosting Linux Ubuntu CI/CD GitHub GitHub Actions Husky Claude Code Cursor AI Model Context Protocol (MCP)
 Microservices Clean Architecture

LOGISTICS & TRANSPORT

07

02/2021 – 05/2023 · 2 YEARS 4 MONTHS · ONGOING MAINTENANCE (SIDE PROJECT)

Ladetrans — Truck Route Management for a Humanitarian Aid Organisation

Role: Sole Developer · Full-Stack & AI **Client:** [CDH Stephanus](https://cdh-stephanus.org)



Team: Solo (sole developer) · client project

Location: Bremen, Germany · Remote

Project Language: German

Website: cdh-stephanus.org

Methodology: Self-directed

RESPONSIBILITIES

- Loading lists per trip: goods to be shipped, recipient addresses and drivers assembled in a single case.
- Every trip as one continuous case — create, edit, close and archive.
- Automated generation of the route documents from the captured data, among other things for border checks.
- Audit-proof archiving of completed trips including the proofs uploaded at the end of each trip.
- Sole responsibility for the backend architecture (**NestJS**, **Clean Architecture**), role-based access control (**better-auth**) and the self-hosted deployment (**Docker**, **Nginx**) including ongoing maintenance (side project).

RESULTS

- According to the client, automated document processes save up to **1,000 hours** per employee per year — about **€ 40,000** annually at an assumed labour cost of € 40 per hour.
- According to the client, up to **30 %** higher logistics efficiency thanks to end-to-end management of the shipping trips.
- Markedly fewer transcription errors; every trip is archived audit-proof together with its documents and proofs.

TECHNOLOGIES

TypeScript Node.js NestJS Next.js React Fastify REST API GraphQL PostgreSQL Redis MikroORM React Router
 Tailwind CSS Single Page Application (SPA) Logistics Route Planning PDF Generation DOCX Generation Digital Archiving better-auth
 JWT Authentication Role-Based Access Control (RBAC) Docker Self-Hosting Linux Ubuntu CI/CD GitHub Actions Husky
 Claude Code Cursor AI Model Context Protocol (MCP) Slack Nginx Microservices Clean Architecture

EDTECH / E-LEARNING

08

05/2020 – 02/2023 · 2 YEARS 10 MONTHS

XU Navigator — Enterprise Learning Platform

Role: Full-Stack Engineer**Client:** XU Group

↗ Recommendation

**Team:** 5–10 people**Location:** Bremen, Germany · Remote**Project Language:** German & English**Website:** xu.de**Methodology:** Scrum

RESPONSIBILITIES

- Co-developed “XU Navigator”, a scalable microservice learning platform for enterprise employees — for clients including **Daimler**, **VW**, and **ThyssenKrupp**.
- Built dynamically, tenant-specific course content (videos, podcasts, multiple-choice quizzes, articles) across clearly scoped services.
- Implemented learning-progress tracking incl. automatic certificate generation as well as an expert and community forum.
- Full-stack development across all platform modules within an agile Scrum team — from course management to the certificate engine, in both frontend and backend.
- Built scalable backend services (**NestJS**, **GraphQL**) with clearly scoped domains in a microservice architecture.
- Built responsive **Angular** interfaces (Angular Material, **NgRx**) focused on reusability and maintainability.
- Deployed via **Azure** (Azure DevOps Pipelines, **Docker**) with automated builds and deployments.

RESULTS

- Scalable microservice platform in production use at major enterprises (**Daimler**, **VW**, **ThyssenKrupp**, and others).
- High degree of customization for tailored, company-specific training programs: every corporation runs its own training on the same platform.
- Automated certificate issuance and progress analytics to boost completion rates.
- Close, cross-functional collaboration with product and design over two and a half years — from initial rollout to production use at multiple enterprise clients.

TECHNOLOGIES

TypeScript Angular Node.js NestJS Express.js REST API GraphQL MongoDB Mongoose NgRx RxJS Angular Material
 SCSS Figma Microsoft Azure Azure DevOps Azure Pipelines Azure App Service Azure Blob Storage Docker CI/CD JWT
 OAuth 2.0 Single Sign-On (SSO) Learning Management System (LMS) Video Streaming Microservices Clean Architecture

HEALTHTECH / MEDICAL DEVICES

09

01/2019 – 05/2020 · 1 YEAR 5 MONTHS

Medical Device Management System

Role: Full-Stack Engineer **Client:** Hospital Group (anonymized)**Team:** 2–4 people**Location:** Germany · On-site / Hybrid**Project Language:** German**Visibility:** HealthTech project for a hospital group (anonymized)**Methodology:** Scrumban

RESPONSIBILITIES

- Co-developed all areas of software to manage medical devices across multiple hospitals — a tool for technicians to simplify maintenance and repair.
- Built device and **master-data management** (hospital, location, device specifications) on top of a cleanly layered architecture (**Clean Architecture, Dependency Injection**).
- Developed a detailed maintenance history (timing and type of work performed) plus device-linked **document management**.
- Implemented fault reporting by nursing staff for fast handoff to technicians.
- Built multi-tenant filtering by hospital (**Multi-Tenancy**) plus integration of Microsoft sign-in via **O365 Identity**.
- Built a type-safe REST API (**ASP.NET Core Web API**, Swagger/**OpenAPI**) with clean layering and **dependency injection**.
- Data modeling and access with **Entity Framework Core** (code-first migrations) on **MSSQL**.
- Automated builds via Azure DevOps Pipelines.

RESULTS

- Delivered complete, multi-tenant device management with maintenance history and **document management**.
- Structured technical foundation for more efficient maintenance and repair processes.
- Contributed for over a year as part of a small, closely aligned team on a safety-critical application in a hospital setting — with a focus on rigor, traceability, and stable delivery.

TECHNOLOGIES

C# .NET Core ASP.NET Core ASP.NET Core Web API OpenAPI Swagger TypeScript React Entity Framework Core Data Modeling
 Microsoft SQL Server (MSSQL) Fluent UI Redux React Router Axios Single Page Application (SPA) Dependency Injection JWT
 O365 Identity Role-Based Access Control (RBAC) Multi-Tenancy Master Data Management Document Management Microsoft Azure
 Azure SQL Azure DevOps Docker CI/CD Clean Architecture